-Project Proposal-

Development of an Enhanced BEMMS for Sri Lankan Hospitals

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Introduction

Biomedical equipment plays a critical role in diagnosis, monitoring, treatment, and patient safety in modern hospitals. Its value depends not only on purchase, but also on how effectively it is maintained, calibrated, tracked, and documented throughout its lifecycle. The World Health Organization notes that a sound maintenance programme should combine inspection, preventive maintenance, corrective maintenance, and record control in order to reduce failures and support safe clinical use.

In many healthcare settings, maintenance information is still handled through fragmented files, manual follow-up, and disconnected records. This weakens service coordination, delays response, and limits managerial visibility. For this reason, digital maintenance platforms are increasingly important, especially when they integrate equipment records, service history, reminders, and performance information within one structured environment.

This project proposes the development of an enhanced Biomedical Equipment Maintenance Management System (BEMMS) for Sri Lankan hospitals. Instead of creating a completely new platform, the project will extend a previously developed BEMMS and strengthen it with automated technician allocation, integrated maintenance and calibration tracking, technician performance monitoring, improved equipment categorisation, and guided fault reporting.

Background and Motivation

Biomedical equipment management is broader than fault repair alone. It includes preventive maintenance, calibration, work assignment, documentation quality, service prioritisation, and monitoring of asset status across departments. Research shows that maintenance systems are more effective when they are planned systematically and supported by reliable information rather than ad hoc paper-based practice.

Studies on healthcare maintenance management indicate that effective maintenance supports better decision-making, reduces downtime, improves reliability, and helps hospitals use scarce technical resources more efficiently. Digital systems are especially useful when they combine inventory control, service history, reminders, and management oversight in one platform.

My earlier BEMMS research addressed the need for a digital maintenance system for Sri Lankan hospitals by introducing structured equipment records, QR-code-based fault reporting, maintenance scheduling, and reminder support. That project demonstrated the feasibility of replacing several manual processes with a more organised software-based workflow.

However, the earlier version remains limited when compared with real biomedical engineering practice. Technician assignment is not automated, calibration follow-up is not fully integrated with maintenance planning, managerial performance visibility is weak, and fault reporting does not yet provide strong guidance to users. I am motivated to undertake this project because I already understand the existing platform and its gaps, which places me in a strong position to extend it in a focused and achievable way.

Problem in Brief

The specific problem addressed in this project is that the previously developed BEMMS does not yet include several operational and managerial functions required for routine hospital use. As a result, technician assignment, calibration follow-up, service supervision, and clear fault reporting may still depend on manual coordination, fragmented records, and delayed monitoring.

This problem is worth solving because maintenance management should not be limited to recording completed repairs. It should also support planning, prioritisation, supervision, documentation quality, and informed decision-making. An enhanced BEMMS that addresses these gaps can improve response coordination, strengthen management visibility, and create greater practical value for Sri Lankan hospitals than a basic maintenance-only system.

Aim

The aim of this project is to develop an enhanced Biomedical Equipment Maintenance Management System for Sri Lankan hospitals by extending the existing BEMMS with automated technician management, integrated maintenance and calibration support, improved equipment organisation, and guided fault reporting using Python, Django, and PostgreSQL.

Objectives

- To critically review the biomedical equipment maintenance management domain, with emphasis on preventive maintenance, calibration control, reliability, and digital maintenance information systems.

- To critically study the software technologies and design approaches suitable for developing a secure, scalable, and user-friendly hospital maintenance management platform.
- To analyse the functional limitations of the existing BEMMS and define the requirements for an enhanced version aligned with real biomedical engineering workflows.
- To design and develop an enhanced BEMMS incorporating automatic technician allocation, technician performance evaluation, integrated maintenance and calibration reminders, department-wise equipment categorisation, and fault suggestion support.
- To evaluate the effectiveness of the enhanced system through functional testing, scenario-based validation, and user-oriented review of workflow relevance.
- To prepare the final documentation, system explanation, and project presentation.

Proposed Solution:- Enhanced BEMMS

The proposed solution is a web-based enhanced version of the existing BEMMS, developed using a three-tier architecture consisting of the user interface, application logic, and database layer. Django will be used as the main web framework and PostgreSQL will manage structured data storage. Role-based access will be applied so that hospital staff, technicians, and managers can interact with the system according to their responsibilities.

The enhanced platform will introduce five major improvements. First, the system will support automatic technician allocation based on equipment category, required skill area, technician availability, and workload. Second, it will include a manager-only technician performance evaluation module that analyses service completion patterns, response time, repeat faults, and pending work. Third, maintenance reminders and calibration reminders will be integrated within one connected scheduling structure rather than handled separately.

Fourth, the system will provide improved department-wise equipment categorisation with clearer asset-level information and status visibility. Fifth, it will offer fault suggestion support so that users reporting a problem can choose guided fault categories and prompts, leading to clearer service tickets for the biomedical engineering unit. Together, these enhancements are intended to improve coordination, documentation quality, supervision, and decision-making.

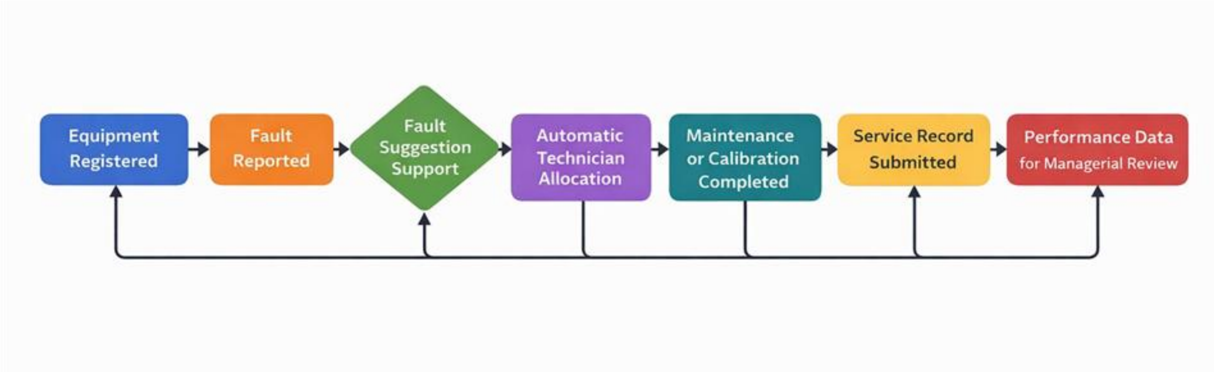

Figure 1 High-level workflow of the proposed enhanced BEMMS

The completed prototype will be evaluated through functional test scenarios and workflow-based validation so that the project demonstrates both technical operation and practical relevance to hospital maintenance activities.

Resource Requirements

The main hardware requirement is a laptop or desktop computer for software development, testing, and documentation. A smartphone will be useful for QR-based workflow testing. Internet access will be required for software packages, development support, and notification testing. The main software requirements are Python, Django, PostgreSQL, Visual Studio Code or an equivalent IDE, a modern web browser, QR code libraries, charting support for dashboards, and Microsoft Word for documentation. Because the project extends an existing prototype and uses mainly open-source tools, the expected development cost is low.

Deliverables

- An enhanced BEMMS software prototype
    - Automatic technician allocation module
    - Technician performance evaluation dashboard
    - Integrated maintenance and calibration reminder module
    - Improved department-wise equipment categorisation interface
    - Fault suggestion support feature
- Final report, technical documentation, workflow diagrams, and testing evidence

Suggested Starting Point

The project will begin with a structured review of the current BEMMS prototype in order to identify reusable components, database dependencies, and functional gaps. This will be followed by a focused literature review on medical equipment maintenance management, calibration tracking, preventive maintenance prioritisation, and digital equipment information systems. After that, the functional requirements of the enhanced version will be finalised and translated into updated system models, database changes, and development milestones.

Project Plan

Time Period	Activity
Week 1	Review existing BEMMS, confirm project scope, and identify functional gaps.
Week 2	Conduct literature review and define detailed system requirements.
Week 3	Redesign database and system architecture for the enhanced modules.
Week 4	Develop automatic technician allocation and task assignment workflow.
Week 5	Develop the integrated maintenance and calibration reminder module.
Week 6	Develop technician performance monitoring and improved equipment categorisation.
Week 7	Implement fault suggestion support and perform testing, debugging, and refinement.
Week 8	Conduct evaluation and complete final documentation and submission materials.

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